AGILE SOFTWARE DEVELOPMENT PROPOSAL & SCOPE SPECIFICATION

PROJECT PROPOSAL & AGILE SCOPE AGREEMENT

Local-First Pre-Order Billing, Cake Customization & Kitchen Production System
Built on Microsoft SQL Server | 3-Phase Agile Delivery | Strict 1-to-1 Scope Definition

■ EXECUTIVE PROJECT OVERVIEW

Client Organization: The Royal Bakery (TRB Systems - Colombo / Outlets)

System Objective: Replace Slow trbsys.net Portal with a Fast Local System on Microsoft SQL Server

Delivery Model: Agile Methodology in 3 Monthly Phases (12 Weeks Total)

Investment Budget: Starting from LKR 400,000 (Baseline Scope — Open for Commercial Negotiation)

Database Engine: Microsoft SQL Server (MSSQL) — Seamlessly Shared with DanuPOS Retail POS

Strict Scope Boundary: Covers Exactly the 15 Core Functional Screens of the Existing System (Zero Unpaid Scope Creep)

Key Performance Guarantee: Sub-50ms Counter Response Time, 100% Offline Billing Resilience, Dual Printing (80mm & A5)

Target Client: The Royal Bakery Management & Executive Operations Team
Document Version: 2.1 — Production-Aligned Agile Scope & Commercial Specification
Date of Submission: September 2026
Prepared By: Lead Systems Engineering Architect & Development Team

TABLE OF CONTENTS

1. Executive Summary & Purpose	Page 3
2. Strict Scope Boundary & Change Request (CR) Policy	Page 4
2.1 Complete Audit Table of the 21 Production System Screenshots	Page 4
3. Exact System Menus & Page Specification (Matching trbsys.net 1-to-1)	Page 6
3.1 Menu 1: Dashboard (/dashboard) [Screenshot #01]	Page 6
3.2 Menu 2: Main Access [Screenshots #02 - #07]	Page 6
• System Users: Create User & Manage Users [Screenshots #02 & #03]	Page 6
• Manage Product / Create Product (/add-product) [Screenshot #04]	Page 7
• Product Categories: Main & Sub Categories [Screenshots #05 & #06]	Page 7
• Customer List (/customer_list - High-Speed Directory) [Screenshot #07]	Page 8
3.3 Menu 3: Orders [Screenshots #08 - #12]	Page 8
• Create Orders (/add-order - Counter Intake & Cake Greetings)	Page 8
• Manage Orders (/manage_orders - Order Grid & Status Badges) [Screenshots #08 & #12]	Page 9
• Make Payment Modal (Counter Balance Settlement)	Page 9
• Cancelled Orders (/cancel_orders - Audit Trail) [Screenshot #09]	Page 10
• Authorize Bills (/bill_authorized - Supervisor Queue) [Screenshots #10 & #11]	Page 10
3.4 Menu 4: Chef Reports (/reports_cheff - 'Order Reports') [Screenshot #13]	Page 10
3.5 Menu 5: Delivery (/delivery - 'Today's Delivery Reports') [Screenshot #21]	Page 11
3.6 Menu 6: Reports (Invoice, Product & Payment Reports) [Screenshots #14 - #19]	Page 11
3.7 Menu 7: SMS Promotions (/Promotions - 'Promotion SMS') [Screenshot #20]	Page 12
3.8 Dual Printing Engine (Direct 80mm Slip & Formal A5 Invoice)	Page 12
4. Agile 3-Phase Delivery Framework (3 Months / 12 Weeks)	Page 13
5. Technical Architecture: Microsoft SQL Server & DanuPOS Integration	Page 15
6. Non-Functional Quality Benchmarks (Speed, Offline, Reliability)	Page 17
7. Commercial Budget, Investment Breakdown & Payment Milestones	Page 18
8. Warranty, Training & Project Sign-Off Agreement	Page 20
Appendix: Table of Architectural Figures & Flowcharts	Page 21

LIST OF FIGURES & ARCHITECTURAL CHARTS

Figure 1	Local-First LAN Architecture (Microsoft SQL Server & DanuPOS Shared Instance)	Page 15

Figure 2	Microsoft SQL Server Relational Entity Relationship Diagram (ERD)	Page 15
Figure 3	End-to-End Pre-Order Billing, Cake Customization & Kitchen Workflow	Page 13

1. EXECUTIVE SUMMARY & PURPOSE

The Royal Bakery (TRB) is one of the most respected bakeries in Sri Lanka. Every day, the bakery takes in hundreds of custom celebration cakes, daily bread orders, and catering orders from walk-in customers, hotels, and schools. Currently, TRB uses a cloud website called trbsys.net for this pre-order billing. However, this system causes severe delays: counter screens freeze, cashiers wait seconds for items to load, and if the showroom internet disconnects, billing stops completely.

This project will replace trbsys.net with a modern, ultra-fast Local-First Counter System running on Microsoft SQL Server. It runs directly on TRB's counter computers, meaning zero internet lag and 100% offline billing reliability. Furthermore, because it uses Microsoft SQL Server, it shares the exact same database engine as TRB's retail POS (DanuPOS), making future integration simple, clean, and unified.

2. STRICT SCOPE BOUNDARY & CHANGE REQUEST (CR) POLICY

To ensure that this project is delivered strictly within the 3-month timeline and budget, it is essential that both The Royal Bakery and the development team have a crystal-clear understanding of the project boundaries.

■ STRICT SCOPE AGREEMENT & ANTI-SCOPE CREEP CLAUSE

1. Exact Feature Copy: The scope of this project is strictly limited to rebuilding the exact functional menus, pages, and features currently present in the production trbsys.net portal as documented across the 21 production screenshots and counter intake workflows. No more, no less.

2. Out-of-Scope Items: Any new capability not present in the current trbsys.net portal (such as e-commerce customer ordering apps, online payment gateways, automated SMS marketing blasts, general accounting ledgers, or attendance systems) is strictly out-of-scope for this contract.

3. Change Request (CR) Process: If TRB requests any new feature, extra screen, or functional change that does not exist in the current system, it must be submitted as a written Change Request (CR). Each CR will receive a separate commercial quotation and timeline, and will only be developed under a separate agreement after the main system launch.

4. Sign-Off Protection: This scope boundary protects both parties against misunderstandings, guarantees on-time delivery, and ensures the highest quality of engineering.

2.1 Complete Audit Table of the 21 Production System Screenshots (from Client PDF)

The client provided an operational walkthrough PDF containing 21 individual screen captures showing every menu, form, and table used across The Royal Bakery. Every single one of these 21 screenshots has been cataloged and verified below:

Screenshot #	PDF Page	Menu / Area	URL Route & Screen Title	Captured Elements & Key Controls
Screenshot #01	Page 1 (Top)	Dashboard	/dashboard Operational Dashboard	Top navbar, KPI cards: Total Orders, Total Pending Orders, Total New Customers (This Month admin & Last Month)
Screenshot #02	Page 1 (Bottom)	Main Access > System Users	/create-users Create User	Main Access menu hover; User Name*, Password*, Confirm Password*, User Type* (Select Level), Submit, Cancel
Screenshot #03	Page 2 (Top)	Main Access > System Users	/manage-users Manage Users	System Users submenu; User list table: User Name, User Role, Action (Edit pen, Delete cross)
Screenshot #04	Page 2 (Bottom)	Main Access > Manage Product	/add-product Create Product	FEED PRODUCT DATA form (Main Cat, Sub Cat, Name, Price, Net Weight, Description, Discount); 850+ Product Table
Screenshot #05	Page 3 (Top)	Main Access > Product Categories	/manage-categories Manage Main Categories	CAT ID#, Main Category, Main Category Description (Chef in charge, e.g. Cheff Premaratna); 8 Categories Table
Screenshot #06	Page 3 (Bottom)	Main Access > Product Categories	/manage-sub-categories Manage Sub Categories	SUB CAT ID#, Main Category dropdown, Sub Category, Sub Category Description; Sub Categories Table
Screenshot #07	Page 4 (Top)	Main Access	/customer_list Customer List	Customer directory table: Customer Name, Mobile, LandLine, Address, Email, Search bar
Screenshot #08	Page 4 (Bottom)	Orders	/manage_orders Manage Orders	Orders menu dropdown (Manage Orders, Cancelled, Authorize); Total Amount badge: 217,550; Orders Table
Screenshot #09	Page 5 (Top)	Orders	/cancel_orders Cancelled Orders	Orders submenu; Cancelled orders audit table: Invoice Id, Reason (CXL), Cancelled By (Admin), Date

Screenshot #10	Page 5 (Bottom)	Orders	/bill_authorized Authorize Bills	Orders submenu; Authorize Bills table: Invoice ID, Name, Delivery Date, Total, Balance, Print-A5, Bill Authorize
Screenshot #11	Page 6 (Top)	Orders	/bill_authorized Authorize Bills (Full View)	Clean full screen view of Authorize Bills queue without dropdown overlay; Pagination controls
Screenshot #12	Page 6 (Bottom)	Orders	/manage_orders Manage Orders (Full View)	Clean full screen view of Manage Orders table; Total Amount badge, Print 80mm & A5, Edit Order, Cancel Order
Screenshot #13	Page 7 (Top)	Chef Reports	/reports_cheff Order Reports	Main-Product-Type*, Sub-Product-Type, Product, Date Range, Print Reoprt, Export; Delivery Date, Product, Description (cake greeting), Qty
Screenshot #14	Page 7 (Bottom)	Reports > By Invoice	/reports_all All Report	Reports menu; By Invoice submenu; Print Reoprt, Export, Bill Authorized; Order list with Make Payment, Status, Due
Screenshot #15	Page 8 (Top)	Reports > By Invoice	/overall-invoices Overall Invoices	By Invoice submenu; Date Range filter (mm/dd/yyyy to mm/dd/yyyy)
Screenshot #16	Page 8 (Bottom)	Reports > By Product	/reports_products_only All Product Sales	By Product submenu; Item Category, Sub Category, Item, Time Range, Search; Table: Main Cat, Sub Cat, Name, Price, Qty, Total Sales
Screenshot #17	Page 9 (Top)	Reports > By Product	/reports-overall Reports Overall (Select)	Customer dropdown selector
Screenshot #18	Page 9 (Bottom)	Reports > By Product	/reports-overall Reports Overall (Filters)	Full filters: Customer, Item Category, Sub Category, Item, Time Range, Payment Method, Search; Print Reoprt, Export
Screenshot #19	Page 10 (Full)	Reports > By Payment	/all-payment Payment Reports	By Payment submenu; Date Range, All Payment Type,

				Search, Print Reoprt, Export; Table: Invoice ID, Cus Name, Payment Method, Paid Amount
Screenshot #20	Page 11 (Top)	SMS Promotions	/Promotions Promotion SMS	PROMOTION SMS template screen: Text area 'Type your message here', Submit, Cancel
Screenshot #21	Page 11 (Bottom)	Delivery	/delivery Today's Delivery Reports	Date of Delivery picker, Search, Print Reoprt; Table: Date of Delivery, Invoice ID, Name and Qty, Cus Name, Address, Phone

3. EXACT SYSTEM MENUS & PAGE SPECIFICATION (MATCHING TRBSYS.NET 1-TO-1)

To make it effortless for Royal Bakery management and counter cashiers to verify the scope, the modules below are organized using the exact menu structure, page names, and field titles found in the production trbsys.net portal, cross-referenced directly to the 21 screenshots verified in Section 2.1.

3.1 Menu 1: Dashboard (/dashboard) [Screenshot #01]

- • **Key Performance Cards:** Page heading matches 'This Month admin' and 'Last Month' cards showing TOTAL ORDERS, TOTAL PENDING ORDERS, and TOTAL NEW CUSTOMERS.
- • **Alerts Banner:** Notification alert banner at the top displaying real-time order intake and status updates.
- • **Navigation:** Navigation sidebar giving instant one-click access to all system menus.

3.2 Menu 2: Main Access [Screenshots #02, #03, #04, #05, #06, #07]

- • **System Users:** Create User form (/create-users) [Screenshot #02] with User Name*, Password*, Confirm Password*, and User Type* (Admin, Accountant, Cashier, Kitchen). Manage Users (/manage-users) [Screenshot #03] table with Edit and Delete options.
- • **Manage Product / Create Product:** Manage Product / Create Product (/add-product) [Screenshot #04] containing Product Main Category, Product Sub Category, Product Name*, Product Price*, Net Weight*, Description*, and Discount* (Discount/Non Discount). Searchable table of all 850+ items with Status filter (Show all, Available, Out of Stock).
- • **Product Categories:** Main Categories (/manage-categories) [Screenshot #05] managing 8 categories (Bread, Cake, Pastry, Kitchen, Restaurant, Shoteats, Showroom) with Chef in charge assignment (e.g., Cheff Premaratna, Cheff Wickramapala). Sub Categories (/manage-sub-categories) [Screenshot #06] managing sub-groups (Buns, Cake Pieces, Drawing Cakes).

- • **Customer List:** Customer List (/customer_list) [Screenshot #07] displaying Customer Name, Mobile, LandLine, Address, and Email. Equipped with high-speed indexing and pagination to eliminate the 60-second loading timeout of the previous system.

3.3 Menu 3: Orders [Screenshots #08, #09, #10, #11, #12]

- • **Create Orders (Customer Profile):** Main showroom intake screen (/add-order). Cashier enters customer mobile number or name for instant auto-fill of Title (Mr/Mrs/Miss/Dr/Rev), First Name, Mobile, Telephone, and Delivery Address. New customers registered in 1 step.
- • **Create Orders (Product Selection):** Cascading selector: Main-Product-Type* -> Sub-Product-Type -> Product (850+ items). Price and net weight load automatically. Cashier enters Quantity (odr_qty) and Item Notes (odr_gre_des).
- • **Create Orders (Cake Customization):** Milestone Greetings* (None, 1st, 2nd, ... 29th) + Occasion Greetings* (None, Happy Birthday, Happy Anniversary, Happy Holy Communion, Congratulations, Happy Valentine's Day) + Custom Icing Name/Text (Greetings info2).
- • **Create Orders (Delivery & Billing):** Date of Delivery (d_delivery), Time (time), Delivery Address (d_address), Ordered By cashier name. Line-item table, Discount %, Service Charge %, Delivery Charge, Gross Total (tot), Advance Paid (odr_adv), Balance Due. Payment type: Cash, Card, Cheque, Staff.
- • **Manage Orders:** Order management grid (/manage_orders) [Screenshots #08 & #12] with Total Amount badge (e.g., LKR 217,550), status filter tabs (Show all, Pending, Paid, Un Paid, Delivered), instant search, and columns: Invoice ID (e.g. RB-0178099), Make Payment, Name & Qty, Date of Delivery, Total, Status, Balance Due, Print (Print 80mm, Print-A5), Actions (Edit Order, Cancel Order).
- • **Make Payment Modal:** Quick pop-up modal triggered by 'Make Payment' button in Manage Orders. Displays Invoice ID, Customer Name, and Balance Due. Cashier inputs payment amount and method (Cash or Card), instantly updating the status to 'Paid / Delivered' and printing receipt.
- • **Cancelled Orders:** Cancelled Orders audit screen (/cancel_orders) [Screenshot #09] logging Invoice Id, Reason (e.g., CXL), Cancelled By username, and Date.
- • **Authorize Bills:** Authorize Bills queue (/bill_authorized) [Screenshots #10 & #11] for supervisor approval of high-value cakes, special catering, or credit orders with Print-A5 and 'Bill Authorize' button.

3.4 Menu 4: Chef Reports (/reports_cheff - 'Order Reports') [Screenshot #13]

- • **Kitchen Filters:** Dedicated kitchen screen for baking chefs. Filterable by Main-Product-Type*, Sub-Product-Type, Product, Date Range (Select two dates), and Status.
- • **Production Grid:** Columns: Delivery Date, Invoice ID, Product Name, Description (containing exact cake greeting & icing inscription), and Quantity. Includes 'Print Reoprt' and 'Export' buttons.

3.5 Menu 5: Delivery (/delivery - 'Today's Delivery Reports') [Screenshot #21]

- • **Dispatch Run Sheet:** Run sheet for delivery dispatchers and drivers. Filter by Date of Delivery, Status, and Search.
- • **Dispatch Columns:** Columns: Date of Delivery, Invoice ID, Name and Qty, Cus Name, Cus Delivery Address, Phone. Includes 'Print Reoprt' button.

3.6 Menu 6: Reports [Screenshots #14, #15, #16, #17, #18, #19]

- • **Reports By Invoice:** All Report (/reports_all) [Screenshot #14] and Overall Invoices (/overall-invoices) [Screenshot #15] filterable by Date Range, Cashier User, Payment Method, and Status. Columns: Invoice ID, Cus Name, Delivery Date, Payment Method, subTotal, Paid Amount, Due, Order Status. Export and Print.
- • **Reports By Product:** All Product Sales (/reports_products_only) [Screenshot #16] and Overall (/reports-overall) [Screenshots #17 & #18] displaying Main Cat, Sub Cat, Product Name, Price, Quantity, and Total Sales across custom date ranges.
- • **Reports By Payment:** Payment Reports (/all-payment) [Screenshot #19] filterable by Date Range and Payment Type (Cash, Card, Cheque, Staff). Shows Invoice ID, Cus Name, Payment Method, and Paid Amount for daily cashier drawer balancing.

3.7 Menu 7: SMS Promotions (/Promotions - 'Promotion SMS') [Screenshot #20]

- • **Promotion SMS:** Promotion SMS drafting screen with text area 'Type your message here' with Submit and Cancel options for promotional message templates.

3.8 Dual Printing Engine (Integrated)

- • **80mm POS Slip:** Instant 80mm thermal receipt printed directly to counter receipt printer for advance deposits and customer pickup vouchers with barcode.
- • **Formal A5 Invoice:** Formal A5 cake invoice containing complete greeting text, cake artwork notes, and terms of collection.

4. AGILE 3-PHASE DELIVERY FRAMEWORK (3 MONTHS / 12 WEEKS)

The project will be delivered using the Agile Methodology, structured into 3 clear monthly phases. At the end of each phase, a live working milestone demonstration will be presented to TRB management for review.

Agile Phase & Period	Deliverable Scope (Exact Menus Covered)	Milestone Review & Sign-Off
PHASE 1 (Month 1) Weeks 1 - 4 Foundation & Master Setup	• Provision Local Master PC with Microsoft SQL Server (MSSQL) • Migrate all 850+ Products, Categories & Customer Database • Menu 1: Dashboard (/dashboard - 'This Month admin') • Menu 2: Main Access complete suite: - System Users: Create User & Manage Users	Milestone 1 Demo & Sign-Off: Live demonstration of Microsoft SQL Server database, 850+ product catalog, and sub-15ms offline search.

	- Manage Product / Create Product (850+ items) - Product Categories: Main & Sub Categories - Customer List (High-speed indexed directory) • Dual Hardware Printer Drivers (80mm & A5)	
PHASE 2 (Month 2) Weeks 5 - 8 Counter Intake & Printing	• Menu 3: Orders core counter intake: - Create Orders (/add-order) with Cake Customization - Milestone Greetings* (1st-29th) & Occasions - Custom Icing Text (Greetings info2) - Advance Deposit & Balance Calculations • Menu 3: Manage Orders (/manage_orders) & Filter Tabs • Menu 3: Make Payment Modal (Balance Settlement) • Direct Hardware Printing Engine (80mm Thermal Slip & A5 Formal Invoice)	Milestone 2 Demo & Sign-Off: Live demonstration of end-to-end counter order creation, cake greeting configuration, and instant receipt printing.
PHASE 3 (Month 3) Weeks 9 - 12 Kitchen, Reports & Launch	• Menu 3: Cancelled Orders (/cancel_orders) & Authorize Bills (/bill_authorized) • Menu 4: Chef Reports (/reports_cheff - 'Order Reports') • Menu 5: Delivery (/delivery - 'Today's Delivery Reports') • Menu 6: Reports (By Invoice, By Product & Payment Reports) • Menu 7: SMS Promotions (/Promotions - 'Promotion SMS') • DanuPOS Shared Microsoft SQL Server Integration Testing • Comprehensive Staff Training (Cashiers, Chefs, Supervisors) & Live Production Cutover	Milestone 3 Demo & Sign-Off: Final User Acceptance Testing (UAT), staff sign-off, and live production deployment at TRB counters.

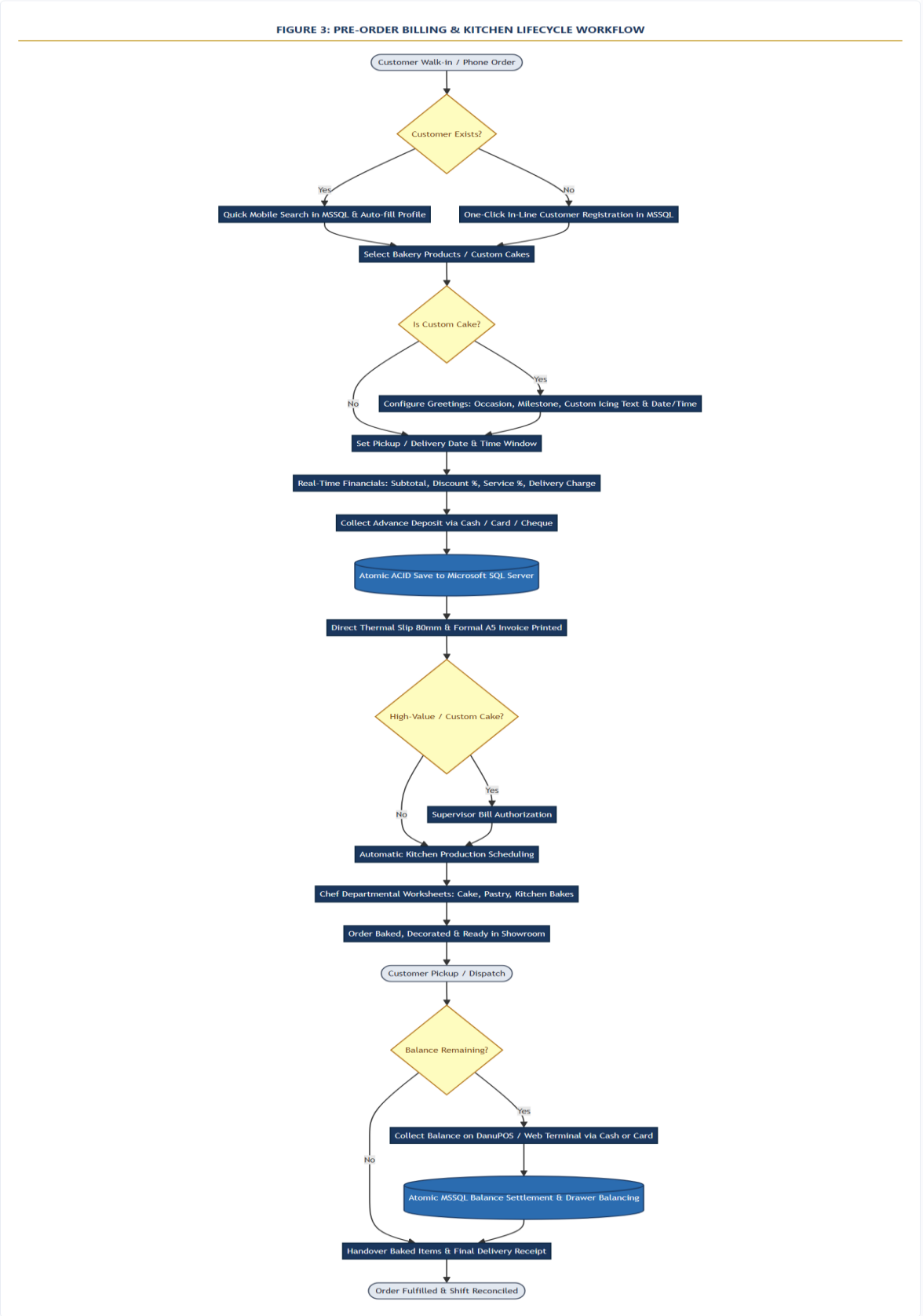


Figure 3: End-to-End Pre-Order Billing, Cake Customization & Kitchen Workflow

Illustrates the complete customer order journey from booking to kitchen baking and final balance settlement.

5. TECHNICAL ARCHITECTURE: MICROSOFT SQL SERVER & DANUPOS INTEGRATION

To ensure the new system operates with zero lag and seamlessly connects with TRB's retail POS (DanuPOS), the system is built on Microsoft SQL Server (MSSQL).

■ WHY MICROSOFT SQL SERVER IS THE STRATEGIC CHOICE FOR TRB

1. Same Engine as DanuPOS: DanuPOS runs on Microsoft SQL Server. Using the same engine allows both systems to live side-by-side on the counter PC with zero compatibility issues.

2. Shared Products & Customers: Pre-orders and DanuPOS retail counters can reference the exact same product and customer records. No duplicate entries or syncing errors.

3. Atomic Balance Settlements: When a customer pays their remaining balance at the DanuPOS counter, a single SQL transaction updates the pre-order status to 'Delivered' and balances the cashier drawer simultaneously.

4. Single Automated Backup: One daily .bak file backup scheduled at night protects both retail POS sales and pre-orders together.

5. Free SQL Server Express Edition: SQL Server Express is 100% free with a 10GB limit (over 10+ years of bakery billing capacity), avoiding expensive software licenses.

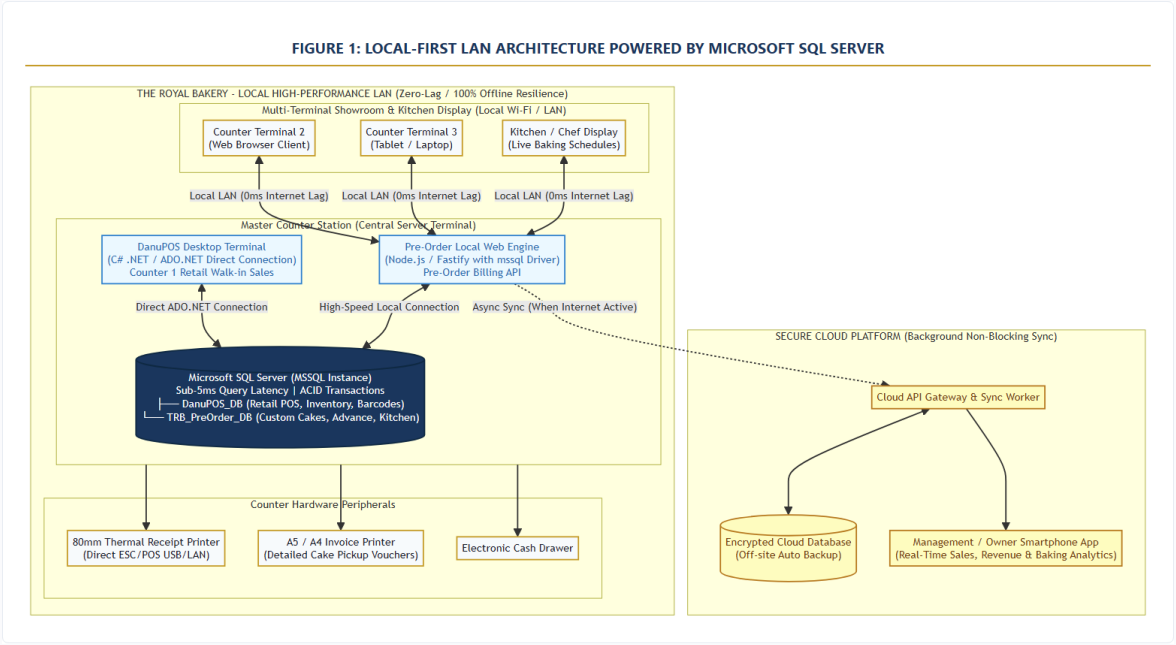


Figure 1: Local-First LAN Architecture (Microsoft SQL Server & DanuPOS Shared Instance)

Shows the Master Counter Station hosting Microsoft SQL Server, serving both DanuPOS and Pre-Order Terminals over the local LAN.

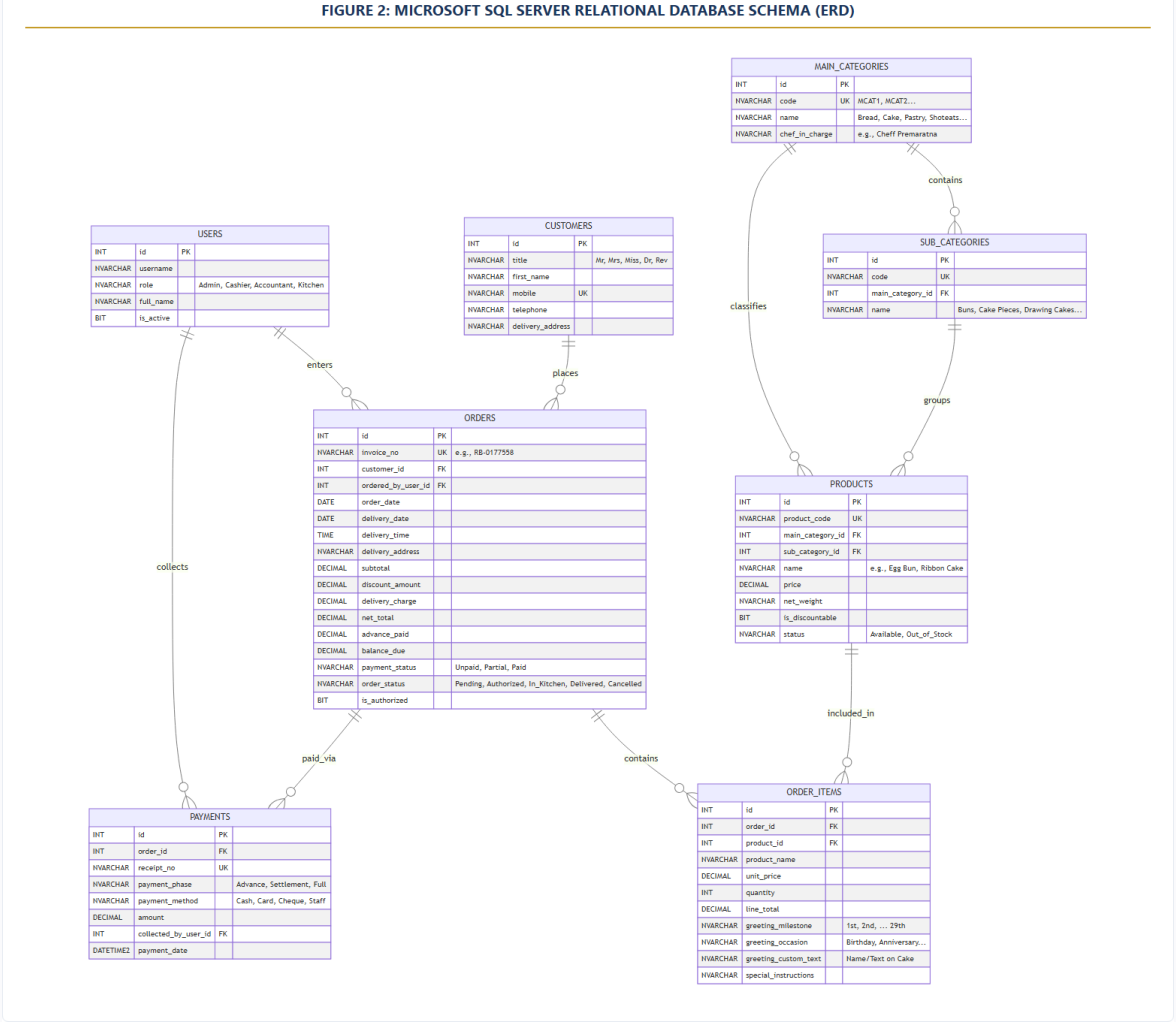


Figure 2: Microsoft SQL Server Relational Database Entity Relationship Diagram (ERD)

Depicts the 8 relational entities: Users, Customers, Main Categories, Sub Categories, Products, Orders, Order Items, and Payments.

6. NON-FUNCTIONAL QUALITY BENCHMARKS

The new system is engineered around strict performance and reliability standards:

Quality Requirement	Target Metric	How It Is Achieved
Counter Speed	Sub-50ms screen response; < 15ms search	Local Fastify API, indexed MSSQL queries, zero internet round-trips
Offline Reliability	100% operational without internet	All billing runs on the local counter PC; internet is never required to bill
Printing Speed	< 1 second print dispatch	Direct raw ESC/POS driver spooling to 80mm printer; no browser preview wait
Data Accuracy	Zero balance discrepancies	ACID compliant SQL Server transactions with automatic rollback on error
Multi-Terminal	Up to 15 counter/kitchen devices	Local Wi-Fi/LAN sharing powered by lightweight Node.js connection pooling

7. COMMERCIAL BUDGET, INVESTMENT BREAKDOWN & PAYMENT MILESTONES

The project investment is structured transparently, starting at a baseline of **LKR 400,000** for the full turnkey scope.

■ COMMERCIAL TERMS & FLEXIBILITY

Baseline Project Investment: LKR 400,000 (Four Hundred Thousand Sri Lankan Rupees)

Negotiable Terms: This price is flexible and open for commercial negotiation based on TRB's finalized counter terminal count and hardware setup.

Complete Turnkey Inclusions: Includes all 15 screens, data migration of 850+ items, dual printer driver setup, DanuPOS shared MSSQL alignment, staff training, and 6 months warranty.

No Recurring Database Fees: Uses free Microsoft SQL Server Express (10GB capacity).

8. WARRANTY, TRAINING & PROJECT SIGN-OFF AGREEMENT

8.1 Comprehensive 6-Month Warranty

Following the live launch, the system includes a 6-Month Free Warranty Period. Any software bugs, database performance issues, or printer driver issues will be fixed immediately at zero extra cost.

8.2 Hands-On Staff Training

- • **Cashiers:** Fast pre-order entry, mobile search, cake customization, advance deposits, and balance collection.
- • **Supervisors:** Order amendments, bill authorization, and cancellation audits.
- • **Chefs:** Accessing daily baking schedules and item summaries.
- • **Accountants:** Daily collection reports and cash drawer balancing.

PROJECT ACCEPTANCE & SCOPE CONFIRMATION

By signing below, both parties agree to the project scope, exact 15-screen deliverables, 3-month Agile timeline, and commercial terms set forth in this document.

For: The Royal Bakery (Client)	For: Systems Development Team
Authorized Signature: _____ Name: _____ Title: _____ Date: _____	Authorized Signature: _____ Name: _____ Title: Lead Systems Engineering Architect Date: _____
Company Seal:	Company Seal: